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3 JULY 1962
SUPERSEDING
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15 December 1959
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9 April 1952

MILITARY SPECIFICATION

INSULATION FELT, THERMAL, GLASS FIBER

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers glass fiber insulation felt for thermal control of machinery and equipment; such as, steam turbines, boilers, boiler feed pumps, and piping at temperature up to 1,200°F.

1.2 Classification. - Insulation felt shall be of the following types, as specified (see 6.2):

Type I - Felted rovings.
Type II - Laminated and felted.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
CCC-T-191 - Textile Test Methods.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.

MILITARY
MIL-P-116 - Preservation, Methods of.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS
C-177-45 - Thermal Conductivity of Materials by Means of the Guarded Hotplate, Method of Test for.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

FSC 5640

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave., at 33rd St. New York 16, N.Y.)

3. REQUIREMENTS

3.1 Material and construction. -

3.1.1 Type I. - The material shall consist of staple glass fibers felted into rovings and woven or bound with wire inserted asbestos thread.

3.1.2 Type II. - The material shall consist of staple glass fibers composed of adhered laminates without use of binders.

3.2 Dimensions and weight. -

3.2.1 Type I. -

3.2.1.1 Length. - Unless otherwise specified (see 6.2), the insulation shall be furnished in rolls 25 feet in length.

3.2.1.2 Width. - Unless otherwise specified (see 6.2), width of roll shall be 60 inches.

3.2.1.3 Thickness and weights. - The insulation shall be furnished in the thicknesses shown in table I as specified (see 6.2), and shall vary not more than plus or minus 10 percent from the weight specified for the ordered thickness.

Table I - Thickness and weights.

Nominal thickness	Thickness tolerance ±	Nominal weight per square foot
Inches	Inch	Ounces
3/4	1/8	9.0
1	1/8	12.0
1-1/2	1/8	18.0

3.2.2 Type II. -

3.2.2.1 Length. - Unless otherwise specified (see 6.2), the insulation shall be furnished in rolls 25 yards in length.

3.2.2.2 Width. - Unless otherwise specified (see 6.2), width of roll shall be 60 inches.

3.2.2.3 Thickness and weights. - The insulation shall be furnished in the thicknesses shown in table II, as specified (see 6.2), and shall vary not more than plus or minus 10 percent from the weight specified for the ordered thickness.

Table II - Thickness and weights.

Nominal thickness	Thickness tolerance ±	Nominal weight per square foot
Inch	Inch	Ounces
1/2	1/8	5.0
3/4	1/8	10.0

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3.3 Thermal conductivity. - Thermal conductivity (k) in B.t.u. per hour per square foot of insulation for 1°F. gradient per inch thickness shall not exceed the values at the mean temperatures shown in table III (see 4.4.3).

Table III - Thermal conductivity.

Mean temperature	Thermal conductivity (k)	
	Type I	Type II
Degrees F.		
300	D.45	0.35
500	56	45
700	.70	.60

3.4 Fineness of fiber. - The diameter of the individual fibers shall average between 0.00030 and 0.00040 inch. The maximum diameter of any fiber shall be not more than 0.00050 inch (see 4.4.4).

3.5 Resistance to vibration. - There shall be no sagging or settling of the insulation when subjected to the vibration test for a period of 100 hours (see 4.4.5).

3.6 Alkalinity. - The alkalinity of the finished material expressed as sodium oxide (Na_2O) shall not exceed 0.20 percent (see 4.4.6).

3.7 Fusing temperature. - The fusing temperature of the fibers shall be not less than 1300°F. (see 4.4.7).

3.8 Stability. - The insulation shall reveal no physical changes upon completion of the stability tests specified in 4.4.8.

3.9 Type II. -

3.9.1 Tensile strength. - The tensile strength before and after heating to 1200°F. shall be not less than 5.0 pounds per square inch (see 4.4.9).

3.10 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - For purposes of sampling a lot shall consist of not more than 25,000 square feet of glass fiber insulation felt of the same type and thickness.

4.2.2 Sampling for acceptance inspection. -

4.2.2.1 Sampling for examination. - A random sample of rolls shall be selected from each lot of material for the examination specified in 4.3.1, with lot acceptance based on table IV.

4.2.2.2 Sampling for tests. - From each lot seven samples, 12 by 60 inches, shall be selected at random for the tests specified in 4.3.2. Each specimen shall be taken from a different roll.

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Table IV - Sampling for examination.

Number of rolls in inspection lot	Number of rolls in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	5	0	1
16 to 40	7	0	1
41 to 65	10	0	1
66 to 110	15	1	2
Over 110	25	2	3

4.3 Examination and tests.-

4.3.1 Examination. - Each of the sample rolls selected in accordance with 4.2.2.1 shall be surface examined, weighed, and measured to determine conformance with the requirements of this specification which do not require tests. Examination shall be conducted as specified in table V. Any roll in the sample containing one or more visual or dimensional defects shall not be offered for delivery, and if the number of defective rolls in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

Table V - Classification of defects in accordance with Standard MIL-STD-105.

Categories	Defects
Critical:	None defined.
Major:	
101	Type of insulation not as specified.
102	Material less than minimum requirements; evidence of unauthorized materials and processes used; insulation not composed of specified materials.
103	Construction of type I insulation nonconforming or not as specified; not woven or bound with wire inserted asbestos thread as specified.
104	Construction of type II insulation nonconforming or not as specified; not adhered laminates as specified; evidence of use of binders.
105	Insulation not furnished in rolls (when applicable); length and width of insulation not as specified.
106	Thickness of insulation not within the specified minimum and maximum tolerances.
107	Evidence of inferior workmanship in the finished product.
Minor:	
201	Packaging, packing and boxing not as specified.
202	Gross weight of containers exceeds the specified maximum.

4.3.2 Acceptance tests. - The samples selected in accordance with 4.2.2.2 shall be subjected to the tests specified in 4.4 to determine conformance with this specification except that the thermal conductivity test (see 4.4.3) shall be conducted only when specified (see 6.2).

4.3.2.1 Action in case of failure. - If any of the samples tested is found to be not in conformance with this specification, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures. -

4.4.1 Thickness. - The test specimen shall be ruled off into 10 approximately square and equal areas, and the thickness measurement taken at the center of each area. In determining the thicknesses, the test specimen shall be placed on a hard flat surface, and the penetrating pin of the depth gage shall be forced downward through the specimen, perpendicular to the flat surface as shown on figure 1. If necessary to

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prevent compression of the specimen by the depth-gage pin, the specimen shall first be pierced. When the point of the pin touches the flat surface, the sliding disk shall be lowered to the point of contact with the top surface of the specimen. The gage shall be withdrawn, and the distance from the point of the pin to the sliding disk shall be measured to the nearest 1/32-inch. The average of the 10 thickness measurements shall be taken as the thickness of the test specimen.

4.4.2 Weight. - Each sample shall be weighed on suitable scale to verify compliance with 3.2.1.3 and 3.2.2.3.

4.4.3 Thermal conductivity. - Thermal conductivity shall be determined in accordance with ASTM Method C-177-45.

4.4.4 Fineness of fibers. - Diameter of fibers shall be determined microscopically on the basis of at least 7 checks on each of the samples. The average diameter for purposes of determining conformance with 3.4 shall be the average of all measurements on all samples; the maximum shall be the maximum diameter of any fiber thus measured.

4.4.5 Resistance to vibration. - The test for determining ability of the material to withstand vibration while subjected to a temperature of 1,200°F. on one surface of the specimen shall be conducted on two 2-foot square, two sheets, 1-inch thick or four sheets, 1/2-inch thick, which shall be mounted on the faces of an electrical heater plate. The ends of the heater plate shall be insulated with cut sections of the material and the entire assembly shall be fitted and mounted within a 1/16-inch thick sheet-iron casing 30 by 30 by 6 inches. The casing shall be mounted in a vertical position on a vibration test apparatus. Five iron constantan thermocouples, equally spaced and secured in each face of the heater plate and the outer surfaces of the metal casing, shall afford a means of ascertaining the inner and the exposed temperature of the assembly. During the test the material shall be subjected to 720 vibrations per minute through an arc of 15 minutes for a period of 100 hours of operation. At the end of the 100-hour period of operation the outer metal casing of the assembly shall be removed and the condition of the sheets noted.

4.4.6 Alkalinity. - Weigh a 5 ± 0.01 gram representative sample of the felt, and introduce into a 500-milliliter pyrex Erlenmeyer flask. Wet with 5 milliliters of 95 percent ethyl alcohol, and add 400 milliliters of distilled water. Reflux for 4 hours ±5 minutes. At the end of this period, disconnect the condenser and filter at once through No. 41 Whatman paper supported in a Buchner funnel. Wash the flask and material three times with 25 milliliters portions of hot distilled water using suction. Titrate immediately with 0.02 Normal H_2SO_4, using 6 to 8 drops of 1 percent solution of phenol-red indicator, to the disappearance of the pink color. Run a blank determination on the same amount of distilled water and alcohol and correct for any alkalinity shown. The percentage alkalinity as Na_2O is calculated from the following formula: percent Na_2O = 0.0124 (mℓ H_2SO_4 used by sample minus mℓ H_2SO_4 used by blank).

4.4.7 Fusing temperature. - Weigh 1 gram of glass fiber into a crucible and place in a muffle furnace at room temperature. Turn all heating elements on at start of test and adjust so that the specified temperature of 1,300°F. is reached in 45 minutes. When this temperature is reached, remove crucible from furnace immediately, allow to cool, and examine visually for fusion. Fusion shall be said to have taken place if any part of the sample has melted and formed a homogeneous mass.

4.4.8 Stability. - Samples of felt 4 inches square shall be encased in a metal wire screen and placed on a rack above the water level in a steam digester. The samples shall be subjected to saturated steam at 225 pounds per square inch gage for 16 hours. Samples shall then be removed, examined, and conditions noted.

4.4.9 Tensile strength. - Tensile strength shall be determined by the grab method described in Method 5100 of Specification CCC-T-191 with the following exceptions: Test specimens shall be 12 by 14 inches in size. The specimens shall be clamped at the top and bottom sections at 1/2-inch pipe. The pipe shall be covered with 1/4-inch thick sponge rubber to prevent cutting and slipping of the specimens. The clamped specimens, having a test area of 1 square foot, shall be attached to the grips of the testing machine. The grips shall separate at a rate of 2 inches per minute until rupture occurs. Tensile strength determinations shall be made on specimens before and after subjection to soaking, heat at 1200°F. for 6 hours. The tensile strength shall be expressed in pounds per square inch (psi) of cross-sectional area.

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4.5 Possible test failures. - Possible test failures are defined as follows:

(a) Weight (Types I and II). -
(1) Weight of insulation not within the specified minimum and maximum values.
(b) Thermal conductivity (Types I and II). -
(1) Thermal conductivity exceeds the allowable maximum when tested as specified.
(c) Fineness of fibers (Types I and II). -
(1) Diameter of any one fiber exceeds the specified maximum; average diameter of the individual fibers not within the specified minimum and maximum values when measured as specified.
(d) Resistance to vibration (Types I and II). -
(1) Evidence of sagging or settling the insulation when tested as specified.
(e) Alkalinity test (Types I and II). -
(1) Alkalinity of the finished material expressed as sodium oxide (Na_2O) exceeds the maximum percentage allowable.
(f) Fusing temperature (Types I and II). -
(1) Fusing temperature of the fibers less than the allowable minimum when tested as specified.
(g) Stability test (Types I and II). -
(1) Evidence of physical changes in the insulation when subjected to the required stability test.
(h) Tensile strength (Type II only). -
(1) Tensile strength, expressed in pounds per square inch (psi) of cross-sectional area, less than the specified minimum when tested as specified.

4.6 Inspection of preparation for delivery. - Sample rolls and shipping containers shall be selected and examined in accordance with Specification MIL-P-116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. - Rolls shall be individually covered with standard 7-1/2 ounce (weight) burlap tubing with tubing drawn together at each end with wire ties.

5.1.2 Level C. - Rolls shall be preserved and packaged in such a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This may conform to the supplier's commercial practice, when such meets the requirements of this level.

5.2 Packing. - Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Type I. -

5.2.1.1 Level A. - Rolls shall be packed in wood-cleated plywood, nailed wood or wirebound boxes conforming to Specification PPP-B-601 (overseas type), PPP-B-621 (class 2) or PPP-B-585 (class 3), respectively, at the option of the contractor. Boxes shall be closed, banded or strapped in accordance with the applicable box specification or appendix thereto. Flat steel strapping, when used, shall be type I, class B. The gross weight of containers shall not exceed 200 pounds.

5.2.1.2 Level B. - Rolls packaged as specified in 5.1.1 need not be overpacked for shipment. Alternatively, rolls may be packed in containers specified for level A (see 5.2.1) except containers shall be of the domestic type or class as applicable. Closure of boxes, when used as shipping containers, shall be as specified in the applicable box specification or appendix thereto.

5.2.1.3 Level C. - Rolls shall be prepared for shipment in a manner which will insure acceptance by common carrier and safe delivery at destination. Containers or method of shipment shall comply with the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.2.2 Type II. - Type II material shall require no overpacking.

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5.3 Marking. - In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - This material is intended for use for thermal control of machinery and equipment; such as, steam turbines, boilers, boilers feed pumps, and piping at temperature up to 1200°F.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Thickness of insulation required (see 3.2.1.3 and 3.2.2.3).
(d) Length and width when other than specified (see 3.2.1 and 3.2.2).
(e) When thermal conductivity tests should be conducted (see 4.3.2).
(f) Levels of packaging and packing required (see 5.1 and 5.2).
(g) Special marking, if required (see 5.3).

Custodians:
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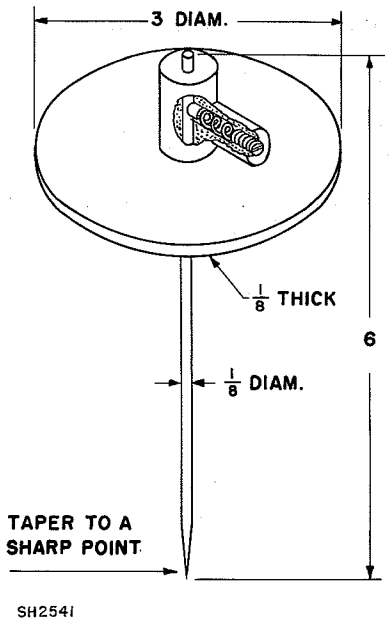


Figure 1 - Depth gage for thickness measurements.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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